INCH-POUND

MIL-C-55365/4D
29 March 1994
SUPERSEDING
MIL-C-55365/4C
23 August 1990

MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLES CWR06 AND CWR09

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55365.

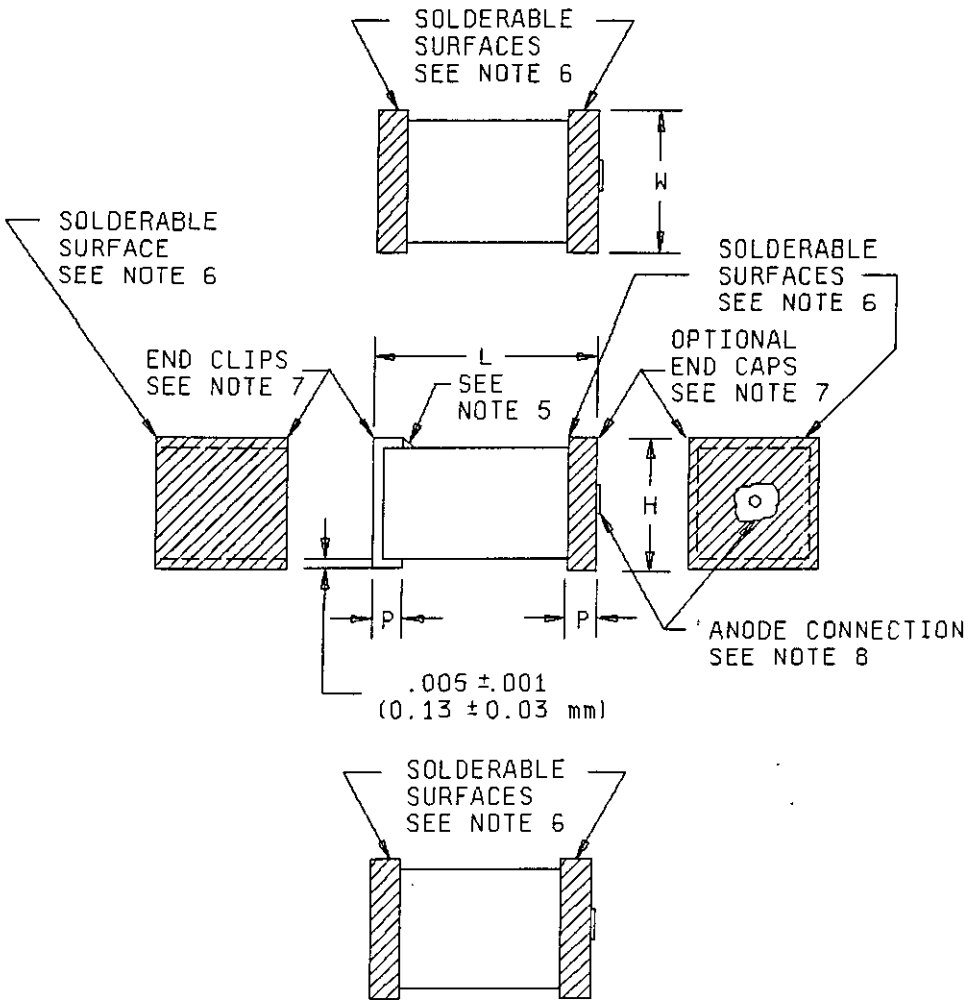
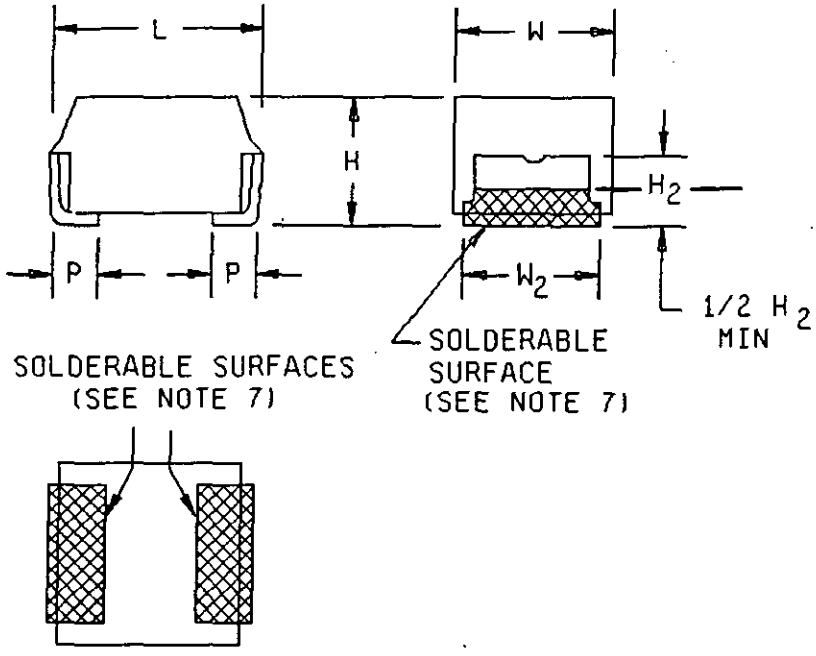


FIGURE 1. Style CWR06 (conformally coated) capacitors.

(D) denotes changes

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Case size	L	W	H	P	W_2	H_2
	±.015 (0.38)	±.015 (0.38)	±.015 (0.38)	±.005 (0.13)		min
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050±.005 (1.27± 0.13)	.030 (0.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050±.005 (1.27± 0.13)	.030 (0.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050±.005 (1.27± 0.13)	.030 (0.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (0.76)	.095+.005, -.010 (2.41+0.13, -0.25)	.030 (0.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (0.76)	.095+.005, -.010 (2.41+0.13, -0.25)	.030 (0.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (0.76)	.130±.005 (.330± 0.13)	.040 (1.02)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)	.105±.005 (2.67± 0.13)	.060 (1.52)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)	.145+.005,-.020 (3.68+0.13, -0.51).	.060 (1.52)

FIGURE 2. Style CWR09 molded chip.

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Step 5 (+125°C): DCL - See table I.
ΔCap. - Shall not exceed ±15 percent of initial +25°C measurement.
DF - See table I.

Step 6 (+25°C): DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial +25°C measurement.
DF - See table I.

Surge voltage: In accordance with MIL-C-55365.

DCL - See table I.
ΔCap. - Within ±5 percent of initial value.
DF - See table I.

(D) Surge current (optional): This test, when applicable, shall be performed on a 100 percent basis before or after the voltage aging test of the qualification and group A exponential tests and before or after the life (accelerated failure rate) test of the group A weibull test. Capacitors shall be subjected to the ten consecutive cycles under the following conditions:

Temperature: Option A: +25°C ±5°C; option B: -55°C ±5°C and +85°C ±5°C.

Applied voltage: Rated dc voltage ±2 percent from a power source having an energy storage bank of 50,000 μF (minimum) across the output terminals.

Charge cycle: 4 seconds ±1 second.

Discharge cycle: 4 seconds ±1 second to a voltage below 1 percent of rated voltage.

The total dc resistance (excluding the capacitor) including the wiring, fixturing, and output impedance of the regulated power supply to each test position during the charging cycle shall be 1 ohm ±0.2 ohm.

On completion of the test, the capacitors shall meet the following requirements:

DCL: See table I.

Cap.: Shall meet the initial measured value.

DF: See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C: DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial measured value.
DF - See table I.

At 85°C: DCL - See table I.

At 125°C: DCL - See table I.

10,000 hours:

At 25°C: DCL - 200 percent of initial requirement.
ΔCap. - Shall not exceed ±10 percent of initial measured value.
DF - See table I.

At 85°C: DCL - 200 percent of initial requirement.

At 125°C: DCL - 200 percent of initial requirement.

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(D) TABLE I. Styles CWR06 and CWR09 characteristics - Continued.

PIN 1/ 2/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C Style CWR06	Max ESR 100 kHz +25°C Style CWR09	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C			
	Volts	µF	µA	µA	µA	%	%	%	Ohms	Ohms	
CWR0-M-474---	35	.47	1.0	10	12	6	8	8	17.0	10.0	B
CWR0-M-684---	35	.68	1.0	10	12	6	8	8	10.0	8.0	C
CWR0-M-105---	35	1.0	1.0	10	12	6	8	8	6.5	6.5	D
CWR0-M-155---	35	1.5	1.0	10	12	6	8	8	4.5	4.5	E
CWR0-M-335---	35	3.3	1.0	10	12	6	8	8	2.5	2.5	F
CWR0-M-475---	35	4.7	2.0	20	24	6	8	8	1.5	1.5	G
CWR0-M-685---	35	6.8	3.0	30	36	6	8	8	1.3	1.3	H
CWR0-N-104---	50	.10	1.0	10	12	6	8	8	75.0	22.0	A
CWR0-N-154---	50	.15	1.0	10	12	6	8	8	25.0	17.0	A
CWR0-N-224---	50	.22	1.0	10	12	6	8	8	17.0	14.0	B
CWR0-N-334---	50	.33	1.0	10	12	6	8	8	12.0	12.0	B
CWR0-N-474---	50	.47	1.0	10	12	6	8	8	8.0	8.0	C
CWR0-N-684---	50	.68	1.0	10	12	6	8	8	7.0	7.0	D
CWR0-N-105---	50	1.0	1.0	10	12	6	8	8	6.0	6.0	E
CWR0-N-155---	50	1.5	1.0	10	12	6	8	8	4.0	4.0	F
CWR0-N-225---	50	2.2	2.0	20	24	6	8	8	2.5	2.5	F
CWR0-N-335---	50	3.3	2.0	20	24	6	8	8	2.0	2.0	G
CWR0-N-475---	50	4.7	3.0	30	36	6	8	8	2.0	1.5	H

1/ Complete PIN shall include additional symbols to indicate style, termination finish, capacitance tolerance, failure rate level, and, if applicable, surge current option letter A (+25°C ±5°C) or B (-55°C ±5°C and +85°C ±5°C). If optional surge current is not required, the last "-" shall be deleted.

2/ Styles CWR06 and CWR09 are, with the acquiring agency approval, considered to be interchangeable, provided the size, capacitance, capacitance tolerance, failure rate level, rated voltage, maximum ESR value, and surge current option (if applicable) remain the same.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99
NASA - NA
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-1866)

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